

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{cases} \partial_t u + u \cdot \nabla u + \nabla P = -\nabla \cdot (\nabla d \odot \nabla d) + \nabla \cdot (\sigma^L(u, d)), \\ \nabla \cdot u = 0, \\ \partial_t d + u \cdot \nabla d - \omega d + \frac{\lambda_2}{\lambda_1} Ad = \frac{1}{|\lambda_1|} (\Delta d + |\nabla d|^2 d) + \frac{\lambda_2}{\lambda_1} (d^T Ad) d. \end{cases}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$(\Gamma^0)_{\underline{\beta}}^{\underline{\alpha}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (\Gamma^1)_{\underline{\beta}}^{\underline{\alpha}} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \Gamma^2 = \Gamma^0 \Gamma^1 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} D &= \sum_{x,y,z \in \mathbb{F}_q^\times} \theta(x^d z) \theta(axz) \theta(bz) \theta(-zy^2) \\ &= \frac{1}{(q-1)^4} \sum_{l,m,n,k=0}^{q-2} G_{-m} G_{-l} G_{-n} G_{-k} T^l(a) T^n(b) T^k(-1) \\ &\quad \times \sum_{x \in \mathbb{F}_q^\times} T^{l+md}(x) \sum_{y \in \mathbb{F}_q^\times} T^{2k}(y) \sum_{z \in \mathbb{F}_q^\times} T^{l+m+n+k}(z). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} g_1(x_1, x_2, y_1) &= 0, \\ \frac{\partial g_1(x_1, x_2, y_1)}{\partial y_1} &= 0, \\ \frac{\partial g_1(x_1, x_2, y_1)}{\partial x_1} f_1(x_1, x_2, y_1) &+ \frac{\partial g_1(x_1, x_2, y_1)}{\partial x_2} f_2(x_1, x_2, y_1) = 0. \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
|\Delta_D^{-\alpha/2} f(x)| &= \left| \int_{\mathbb{R}^N} K_\alpha(x, y) f(y) d\omega(y) \right| \\
&\leq \int_{d(x, y) \leq s} \frac{d(x, y)^{2+\alpha}}{\|x - y\|^2} \frac{1}{\omega(B(x, d(x, y)))} |f(y)| d\omega(y) \\
&\quad + \int_{d(x, y) > s} \frac{d(x, y)^{2+\alpha}}{\|x - y\|^2} \frac{1}{\omega(B(x, d(x, y)))} |f(y)| d\omega(y) \\
&=: J_1 + J_2.
\end{aligned}$$